

## BIOCHEMISTRY

## Ubiquitin receptor–mediated, ubiquitin-independent targeted protein degradation via 26S proteasomes

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The 26S proteasome engages with ubiquitinated substrates primarily through its constituent ubiquitin (Ub) receptors, which initiates a cascade of proteolytic processes. Leveraging this recognition mechanism, we developed a targeted protein degradation (TPD) strategy that recruits substrates directly to the proteasome, thereby bypassing the ubiquitination step. Our proteasome-targeting chimera, Protea-Tac, is a heterobifunctional protein degrader composed of a Ub receptor and an intracellular antibody. This chimera integrates into 26S proteasomes without altering their structural or functional integrity. Protea-Tac with cognate antibodies degraded various target proteins, including c-Fos, BRD4, <sup>Flag</sup>TDP43, <sup>HA</sup>tau, and <sup>GFP</sup>ODC. We mechanistically demonstrated that this platform is (i) modular, allowing facile target switching, (ii) Ub independent, and (iii) highly target specific. Furthermore, Protea-Tac exhibited potent in vivo antitumor efficacy, posttranslationally inducing c-Fos degradation and substantially delaying tumor progression through both viral and nonviral delivery systems. These findings identify Protea-Tac as a distinct TPD platform capable of directly degrading intracellular proteins via engineered 26S proteasomes.

## INTRODUCTION

Targeted protein degradation (TPD) has evolved from a conceptual framework to an emerging therapeutic modality that selectively eliminates disease-related proteins (1). Unlike conventional small-molecule drugs that typically rely on high-affinity binding, TPD can accomplish a catalytic mechanism that allows for effective protein turnover at moderate-to-low binding affinity (2, 3). Proteolysis-targeting chimera (PROTAC) is at the forefront of TPD therapeutics. It usually is a synthetic heterobifunctional molecule composed of a target-binding ligand and an E3 ubiquitin (Ub) ligase-binding ligand. The induced proximity via PROTAC promotes substrate polyubiquitination (hereafter referred to as ubiquitination) and subsequent hydrolysis by 26S proteasomes (4–8). Many other TPD platforms, including molecular glue degraders (MGDs), use a similar mode of action to PROTACs (9–11). These approaches offer distinct advantages over traditional therapeutics, particularly in targeting proteins previously considered undruggable, such as scaffold proteins and transcription factors (12, 13). In addition, TPD has the potential to provide a unique therapeutic solution for acquired drug resistance, which is often caused by mutations in neoplastic targets (14–17).

The mechanistic foundation of these TPD modalities lies in their ability to exploit the natural Ub-proteasome system (UPS), the primary proteolytic pathway in eukaryotic cells (18). In the UPS, E3 Ub ligases, which have over 600 members encoded in the human genome, serve as the primary determinants of target specificity. These enzymes directly interact with to-be-degraded substrates and promote the transfer of charged Ub moieties from cognate E2 Ub-conjugating enzymes. Both PROTACs and MGDs aim to emulate this biochemical process by inducing the formation of an artificial ternary complex that brings the neo-substrate and E3 ligase into favorable spatial proximity and orientation. Despite substantial progress, the identification of suitable ligands for E3 ligases remains challenging; at present, cereblon (CRBN) and von Hippel–Lindau (VHL), account for most of reported PROTACs and MGDs (19–21). Similarly, proteins lacking well-defined hydrophobic pockets or having intrinsically disordered regions are still difficult to target through TPD strategies. As a result, degrader discovery often requires multiple iterative cycles of molecular design, chemical synthesis, functional evaluation, and further optimization.

The 26S proteasome holoenzyme catalyzes the irreversible protein hydrolysis of most short-lived proteins in cells (22). Given its high cellular abundance in most cell types, the proteasome is an ideal degradation machinery for TPD (23–26). The 26S proteasome is composed of two distinct subcomplexes: the 20S catalytic particle (CP) and the 19S regulatory particle (RP), which can reversibly associate and dissociate in response to cellular stress and signaling cues (27–29). These catalytic and regulatory features exemplify an evolutionary dichotomy: the ancient CP for proteolysis and the later-evolved RP for Ub recognition and processing (30). Ub chains are recognized by Ub receptors (UbRs) located in the RP, including PSMD2/Rpn1, PSMD4/Rpn10, and ADRM1/Rpn13. Their interaction initiates a cascade of coordinated processes, including conformational remodeling of the RP, which allows efficient substrate unfolding and translocation (31–33). When the substrate's unstructured region engages with the pore of the PSMC/Rpt adenosine

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triphosphatase (ATPase) ring, nucleotide hydrolysis generates mechanical force, simultaneously unfolding the substrate and translocating it into the hydrolytic chamber (34). The RP also deubiquitinates the substrate to facilitate this translocation process and prevent the collateral Ub degradation (35).

To circumvent the mechanistic limitations associated with E3 ligase-dependent and ubiquitination-inducing TPD strategies, alternative modalities, some of which co-opt the lysosomal or autophagic pathways, have emerged (36–40). In parallel, Ub-independent target degradation has largely been achieved by proximity-driven substrate recruitment to the 26S proteasome as initially demonstrated using inducible recruitment systems and engineered degrons (41–43). Such substrate tethering is not restricted to artificial settings: Pathogen effectors and endogenous cofactors could engage substrates to the proteasome to promote proteolysis bypassing ubiquitination (44, 45). More recently, compounds that recruit neo-substrates directly to UbRs or other RP components have demonstrated the therapeutic potential of Ub-independent TPD (46–50). Despite these advances, the broader application of such small-molecule degraders as a platform technology remains constrained by the need for suitable target-binding ligands and often requires substantial medicinal chemistry optimization to achieve sufficient potency, selectivity, and pharmacological properties. Genetically encoded degraders may provide a complementary and potentially more versatile strategy for expanding the repertoire of degradable targets, although its clinical translation faces important challenges, including efficient delivery and the risk of immunogenicity (51, 52).

In this study, we exploited the biochemical feature of the proteasome-residing UbRs to develop a genetically encoded, directly 26S proteasome-targeting chimera (Protea-Tac). By fusing PSMD4 or ADRM1 to a substrate-targeting antibody [tAb; single-chain variable fragment (scFv) or nanobody (VHH)], we created a modular degrader system that integrates into 26S proteasomes while preserving their proteolytic activity. Biochemical analyses demonstrated that Protea-Tac is modular and heterobifunctional and can induce Ub-independent TPD. Tandem mass tag (TMT)-based quantitative proteomics revealed its high selectivity. In addition, *in vivo* experiments using adeno-associated virus (AAV) and lipid nanoparticles (LNPs) showed that Protea-Tac degraders posttranslationally reduced target proteins and inhibited tumor growth across multiple mouse tumor models. This study introduces Protea-Tac as a distinct, Ub-independent TPD platform, broadening the therapeutic potential of TPD.

## RESULTS

### Protea-Tac as a TPD modality that directly engages the 26S proteasome

On the basis of previous studies, which revealed low micromolar affinity interactions between the UbRs and Ub chains, we hypothesized that intracellular antibodies such as scFv and VHH could serve as alternatives to Ub chain-mediated substrate recruitment to the 26S proteasome (53, 54). We designed the initial Protea-Tac platform by fusing a UbR to an scFv targeting the *c-Fos* oncoprotein, scFv(*c-Fos*), at either the N or C terminus, so as not to sterically impede the 19S subcomplex assembly (Fig. 1A). To assess TPD activity, various Protea-Tac constructs were transiently expressed in A549 cells. We found that PSMD4-scFv(*c-Fos*) significantly lowered the steady-state levels of cotransfected *c-Fos* in an expression level-dependent manner (Fig. 1B). In sharp contrast, *c-Fos* mRNA levels

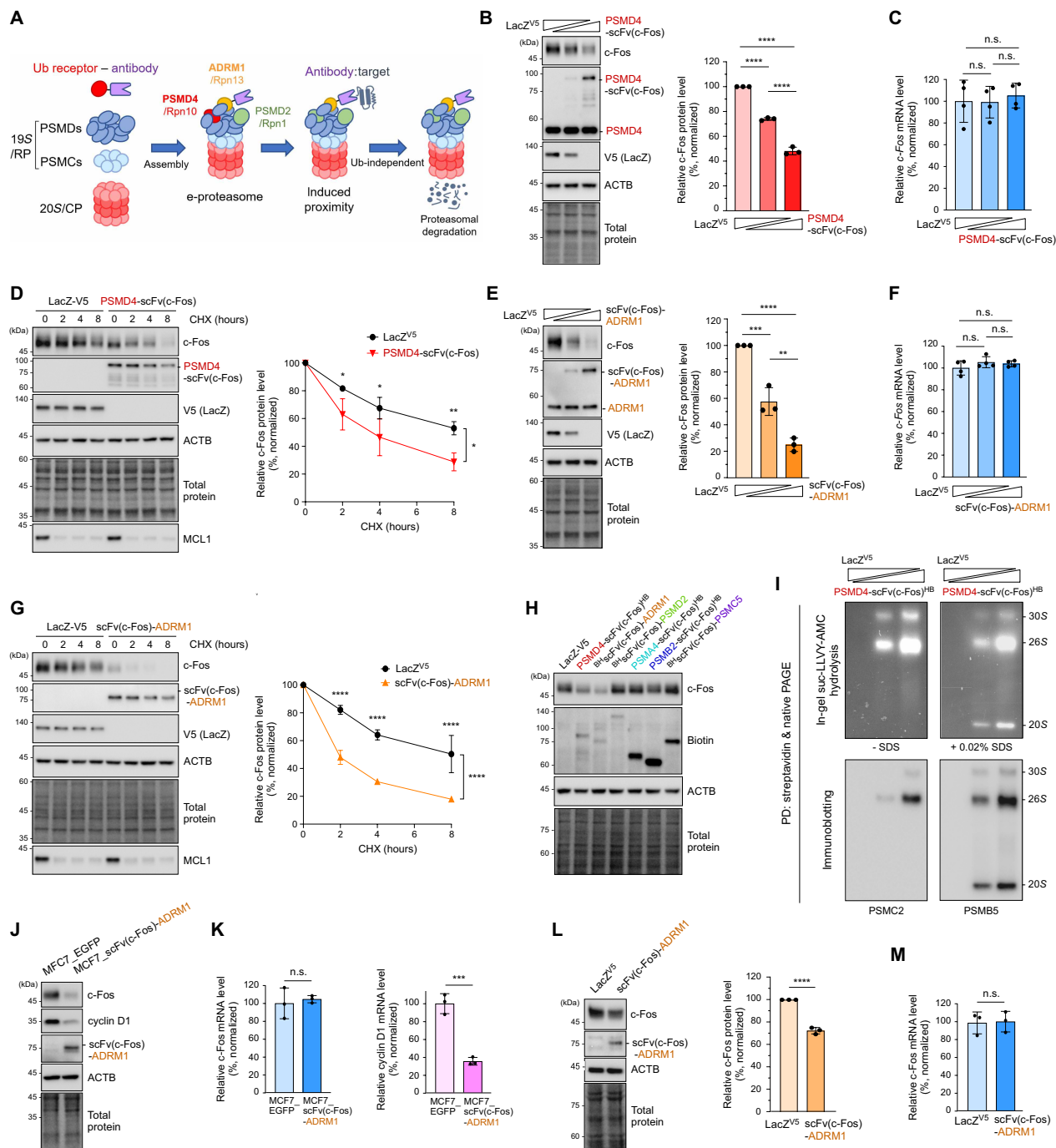
remained unchanged regardless of coexpression of PSMD4-scFv(*c-Fos*), LacZ<sup>V5</sup>, or their combination (Fig. 1C), implying that the reduction in *c-Fos* protein occurred posttranslationally. Time-course chase experiments demonstrated accelerated *c-Fos* degradation in the presence of its cognate, PSMD4-based Protea-Tac chimera (Fig. 1D).

The Protea-Tac degraders with ADRM1 were likewise effective; both ADRM1-scFv(*c-Fos*) and scFv(*c-Fos*)-ADRM1 substantially reduced *c-Fos* protein levels without altering *c-Fos* mRNA (Fig. 1, E and F, and fig. S1, A and B). We found that ADRM1-based Protea-Tacs were more effective in *c-Fos* degradation than PSMD4-based degraders (Fig. 1G and fig. S1C). Notably, fusions of scFv(*c-Fos*) with PSMD2 or other structural components (either CP or RP subunits) of the 26S proteasome did not markedly affect *c-Fos* levels (Fig. 1H and fig. S1, D to G). To assess whether and to what extent the 26S proteasome pool is remodeled through Protea-Tac incorporation, we compared the levels of the incorporated chimeras with those of their endogenous UbR counterparts by immunoblotting (IB) of affinity purified proteasomes. Both PSMD4-scFv(*c-Fos*) and scFv(*c-Fos*)-ADRM1 were detected, albeit at substantially lower levels than endogenous PSMD4 and ADRM1, respectively, suggesting that Protea-Tac is incorporated into only a minor fraction of cellular 26S proteasomes (fig. S1, H and I). We next performed streptavidin-affinity purification using biotin-hexahistidine-tagged Protea-Tac constructs, revealing structurally intact and enzymatically active 26S proteasomes through nondenaturing (native) electrophoresis and in-gel fluorogenic substrate (suc-LLVY-AMC) hydrolysis (Fig. 1I and fig. S1, J to L). Protea-Tac degraders with a mutant UbR that is incapable of direct proteasomal interaction did not show any TPD activity (fig. S1M). Collectively, these results show that direct substrate tethering to the 26S proteasome can trigger TPD when mediated by permissive UbR configurations, such as PSMD4- and ADRM1-based Protea-Tacs.

Having validated Protea-Tac functionality using transient transfection, we next generated stable cell lines overexpressing scFv(*c-Fos*)-ADRM1 in human breast cancer (MCF7) and colorectal cancer (HCT116) cells, where *c-Fos* up-regulation is implicated in tumorigenesis (55, 56). The selected clones expressing the greatest amounts of scFv(*c-Fos*)-ADRM1 exhibited marked endogenous *c-Fos* degradation (72.9 and 58.6% in MCF7 and HCT116, respectively) compared to control cells expressing enhanced green fluorescent protein (EGFP) (Fig. 1J and fig. S1N). Again, the mRNA levels of endogenous *c-Fos* remained unaffected (Fig. 1K). It was noticeable that protein and mRNA levels of cyclin D1, a direct transcriptional target of *c-Fos*, were substantially reduced in these stable cells (Fig. 1, J and K, and fig. S1O) (57, 58). Transient expression of scFv(*c-Fos*)-ADRM1 in MCF7 and HCT116 cells also reduced endogenous *c-Fos* levels but to a lesser extent than when stably overexpressed (Fig. 1, L and M, and fig. S1, P and Q), likely owing to higher proportion of Protea-Tac-expressing cells and sustained expression across the stable cell populations.

### Mechanistic exploration of Ub-independent, proteasomal degradation by Protea-Tac

To understand the mechanistic basis of Protea-Tac-mediated TPD, we first investigated whether both modules, the tAb and UbR, are necessary and whether their fusion is essential. A panel of control constructs expressing only scFv(*c-Fos*), UbR alone (PSMD4 or ADRM1), or their coexpression did not affect cellular *c-Fos* levels, whereas the chimeric Protea-Tac constructs effectively degraded

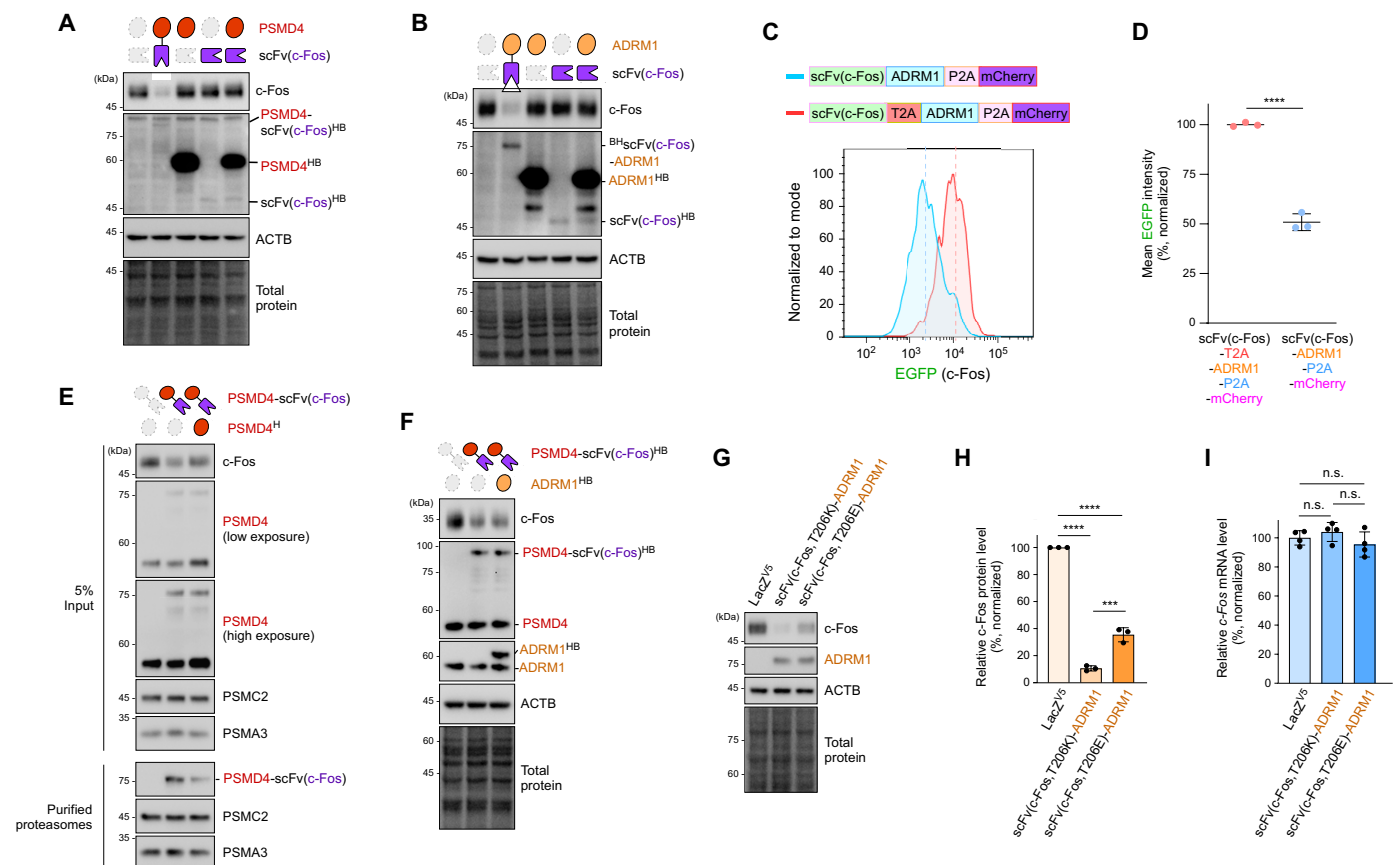


**Fig. 1. The Protea-Tac system enables engineered degradation of target proteins through direct proteasome engagement.** (A) Schematic illustration of TPD by Protea-Tac. (B) A549 cells were transfected with c-Fos and PSMD4-scFv(c-Fos) or LacZ<sup>V5</sup> for 64 hours; WCLs were analyzed by SDS-PAGE/IB and total protein staining. Total plasmid DNA amounts and volumes remained constant across all experimental conditions. Left: Representative IBs from three independent experiments. Right: c-Fos quantification. Protein quantification was normalized to total protein unless indicated otherwise. N = 3, means ± SD, \*\*\*\*P < 0.0001; one-way ANOVA with Tukey's post hoc test. (C) c-Fos mRNA was measured by qRT-PCR and normalized to GAPDH. N = 4, means ± SD, n.s., not significant; one-way ANOVA with Tukey's post hoc test. (D) Cycloheximide (CHX) chase after transfection as in (B), with c-Fos normalized to 0 hours in each group. MCL1 served as a physiological proteasome substrate. N = 3, means ± SD, \*P < 0.05, \*\*P < 0.01; two-way ANOVA with Sidak's post hoc test. (E to G) scFv(c-Fos)-ADRM1 was evaluated as in (B) to (D). (H) Proteasome subunit-based chimera scFv(c-Fos)-ADRM1 was evaluated as in (B) to (D). (I) Hexahistidine-biotin (HB)-tagged PSMD4-scFv(c-Fos) was purified by streptavidin affinity purification and analyzed by native PAGE, in-gel suc-LLVY-AMC hydrolysis, and IB; 0.02% SDS activated 20S proteasomes. (J) Endogenous c-Fos and cyclin D1 proteins were compared in MCF7 cells stably expressing EGFP or scFv(c-Fos)-ADRM1. (K) c-Fos and cyclin D1 mRNAs were measured by qRT-PCR and normalized to GAPDH. N = 3, means ± SD, \*\*\*P < 0.001, n.s.; two-tailed Student's t test. (L and M) Transient scFv(c-Fos)-ADRM1 expression in MCF7 cells was assessed by IB and qRT-PCR. N = 3, means ± SD, \*\*\*\*P < 0.0001, n.s.; two-tailed Student's t test. See fig. S1.

c-Fos (Fig. 2, A and B). We also successfully achieved TPD using VHH as the tAb module, which also demonstrated that both modules are necessary and should be fused for degradation (fig. S2, A and B). To further investigate this heterobifunctionality, we generated constructs that fused scFv(c-Fos) and ADRM1 with or without a self-cleaving T2A sequence at their junction, followed by a P2A-mCherry reporter (Fig. 2C). Flow cytometry analysis of 293 T cells stably expressing EGFP-tagged c-Fos (EGFP-c-Fos) indicated significantly lower EGFP signals in the mCherry-positive cell population that expresses the fused scFv(c-Fos)-ADRM1 than those expressing the T2A-separated format (Fig. 2D).

To further dissect the mechanism and functionality of Protea-Tac, we coexpressed free PSMD4 or ADRM1 alongside Protea-Tac degraders. Coexpression of the cognate UbR competitively reduced the incorporation of Protea-Tac into 26S proteasomes and attenuated its TPD activity (Fig. 2E and fig. S2C). In contrast, excess PSMD4 or ADRM1 had no effect on Protea-Tacs with noncognate

UbRs (Fig. 2F and fig. S2D), indicating that Protea-Tac-mediated degradation is primarily associated with its incorporation into the 26S proteasome holoenzyme rather than with stand-alone UbRs. Similarly, coexpression of cognate free UbRs attenuated the TPD activity of VHH-based degraders (fig. S2, E and F). Protea-Tac chimeras with noncognate antibodies did not affect c-Fos levels (fig. S2, G and H). When we substituted the tAb's CDR region with that of IgG, Protea-Tac activity was completely abolished (fig. S2, I to K). We then investigated the impact of tAb affinity on degradation efficiency, using scFv(c-Fos) variants with varying affinities for recombinant c-Fos (fig. S2L). We found that the lower-affinity Thr<sup>206</sup>→Lys (T206K) mutant improved posttranslational c-Fos degradation, whereas the higher-affinity Thr<sup>206</sup>→Glu (T206E) variant was associated with greater retention of c-Fos in purified proteasomes and induced less efficient degradation (Fig. 2, G to I, and fig. S2M). These findings imply that Protea-Tac degradation efficiency is not simply proportional to tAb binding affinity and that



**Fig. 2. Protea-Tac degraders require both UbR and tAb modules.** (A and B) A549 cells were transfected with c-Fos and PSMD4- or ADRM1-based components expressed either separately or as UbR-scFv(c-Fos) fusions; WCLs harvested at 64 hours were analyzed by SDS-PAGE/IB and total protein staining. Protein quantification was normalized to total protein unless indicated otherwise. (C) Bicistronic and tricistronic constructs encoding scFv(c-Fos)-ADRM1 without or with a T2A sequence at the tAb-UbR junction, followed by P2A-mCherry, were expressed in HEK293T cells stably expressing EGFP-c-Fos and analyzed by flow cytometry. Histograms show EGFP-c-Fos fluorescence in mCherry-positive cells. (D) EGFP mean fluorescence intensity (MFI) in mCherry-positive cells.  $N = 3$ , means  $\pm$  SD, \*\*\*\* $P < 0.0001$ ; two-tailed Student's  $t$  test. (E) HEK293 cells stably expressing HB-tagged PSMB2 were transfected as in (A) with untagged PSMD4-scFv(c-Fos) and PSMD4<sup>H</sup> to assess competitive proteasome incorporation and target degradation. Affinity-purified proteasomes and WCLs were analyzed by denaturing SDS-PAGE/IB for c-Fos and proteasome subunits. (F) PSMD4-scFv(c-Fos) was coexpressed with noncognate ADRM1 and assessed for c-Fos degradation. (G to I) Lower-affinity T206K and higher-affinity T206E scFv(c-Fos) variants were evaluated in ADRM1-based degraders. (G) Representative IBs from three independent experiments. (H) c-Fos protein quantification.  $N = 3$ , means  $\pm$  SD, \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , \*\*\*\* $P < 0.0001$ ; one-way ANOVA with Tukey's multiple comparison test. (I) c-Fos mRNA measured by qRT-PCR and normalized to GAPDH.  $N = 4$ , means  $\pm$  SD, n.s.; one-way ANOVA with Tukey's post hoc test. See fig. S2.

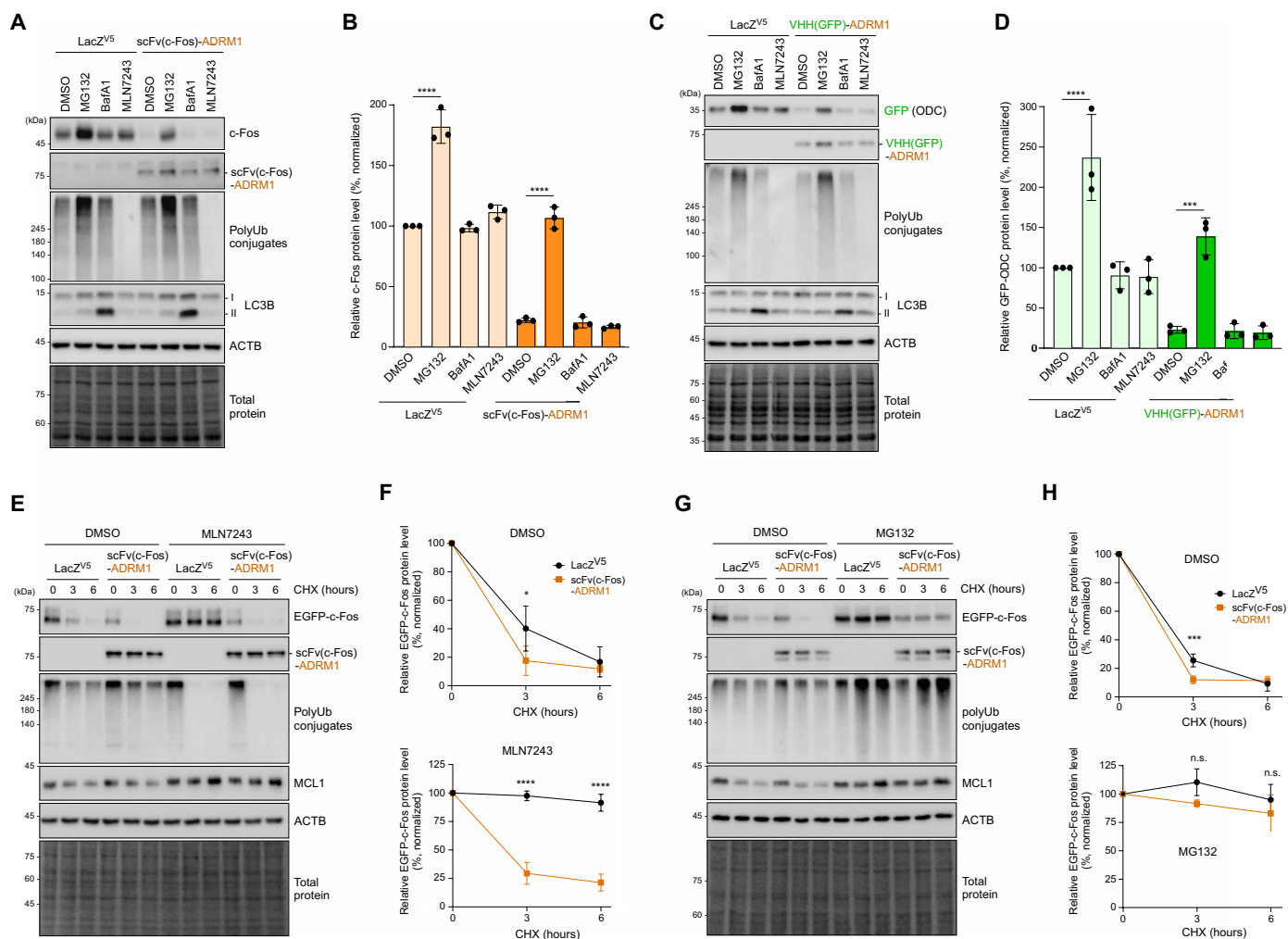
maximal affinity may not necessarily be optimal for productive target turnover.

Because Protea-Tac relies on direct substrate localization to the 26S proteasome rather than ubiquitination, we next sought to investigate its dependency on substrate ubiquitination and proteasomal degradation. The induced degradation of c-Fos and EGFP-c-Fos by PSMD4- or ADRM1-based Protea-Tacs was strongly abrogated by MG132 (a proteasome inhibitor) but not by bafilomycin A1 (an autophagic flux inhibitor) or MLN7243 (an E1 Ub-activating enzyme inhibitor) (Fig. 3, A and B, and fig. S3, A to D). Similarly, VHH(GFP)-ADRM1-mediated degradation of GFP-tagged ornithine decarboxylase (ODC) was sensitive to MG132 but not to MLN7243 (Fig. 3, C and D). In chase experiments, treatment with MLN7243 had little

effect on EGFP-c-Fos turnover by Protea-Tac but MG132 blocked it (Fig. 3, E to H). Flow cytometry analysis of EGFP-c-Fos stable cells further showed that MG132, but not MLN7243, attenuated Protea-Tac-mediated target degradation (fig. S3E). Both PSMD4- and ADRM1-based Protea-Tacs substantially reduced the protein levels of a lysine-less c-Fos mutant [c-Fos (K0)] without altering its mRNA levels (fig. S3, F to H). Together, these findings support a proteasome-dependent, ubiquitination-independent mode of Protea-Tac-mediated degradation.

### Versatility and specificity of modular Protea-Tac degraders

To assess the versatility of the Protea-Tac system in targeting diverse neo-substrates, as well as its modular flexibility allowing component



**Fig. 3. The Protea-Tac system enables Ub-independent and proteasome-mediated TPD.** (A) scFv(c-Fos)-ADRM1-mediated c-Fos degradation was examined in the presence of dimethyl sulfoxide (DMSO), the proteasome inhibitor MG132 (10  $\mu$ M), the vacuolar type H<sup>+</sup>-ATPase inhibitor bafilomycin A1 (BafA1; 200 nM), or the E1 Ub-activating enzyme inhibitor MLN7243 (1  $\mu$ M) for 6 hours. (B) The steady-state levels of c-Fos detected across conditions in (A) were quantified and normalized to total protein. Each bar indicates the means  $\pm$  SD ( $N = 3$ ). \*\*\*\* $P < 0.0001$  (one-way ANOVA followed by Tukey's multiple comparison test). (C and D) As in (A) and (B), except that VHH(GFP)-ADRM1 was used as a degrader and GFP-ODC was a neo-substrate. (E) Chase experiments were performed to track the degradation kinetics of EGFP-tagged version of c-Fos (EGFP-c-Fos) in the presence of DMSO or MLN7243. (F) Quantification of EGFP-c-Fos band intensities in (E), normalized to those of total protein and to the values at 0 hours post-CHX treatment (set as 100%), under DMSO (top) or MLN7243 (bottom) treatment conditions. Average percentages of remaining EGFP-c-Fos (means  $\pm$  SD) from three independent experiments ( $N = 3$ ); \* $P < 0.05$ , \*\*\*\* $P < 0.0001$  (two-way ANOVA followed by Sidak's post hoc test). (G and H) Chase experiments and quantification using EGFP-c-Fos as in (E) and (F) except that they were performed in the presence of either DMSO or MG132. Data represent the means  $\pm$  SD from three independent experiments ( $N = 3$ ). \*\*\* $P < 0.001$  (two-way ANOVA followed by Sidak's post hoc test). n.s., not significant. See fig. S3.

substitution, we first replaced the tAb module with scFvs or VHHs that recognize common epitopes such as Flag, HA, or GFP tags (Fig. 4A). When expressed in A549 cells, both PSMD4-scFv(Flag) and scFv(HA)-ADRM1 degraders effectively reduced the levels of coexpressed <sup>Flag</sup>TDP43 and <sup>HA</sup>Tau proteins, respectively, in both the radioimmunoprecipitation assay (RIPA)-soluble and RIPA-insoluble fractions, whereas there was no change in their mRNA levels (Fig. 4, B to E, and fig. S4, A and B). To assess the effects of Protea-Tac on distinct biochemical pools of tau, we analyzed human embryonic kidney (HEK) 293 cells stably expressing <sup>HA</sup>tau (ON4R) by RIPA fractionation. Although scFv(HA)-ADRM1 expression substantially reduced RIPA-soluble monomeric tau levels, it had little effect on high-molecular-weight (HMW) and insoluble tau species. These results suggest that Protea-Tac primarily reduces the soluble, monomeric tau pool under the conditions tested, rather than directly clearing soluble HMW or RIPA-insoluble tau species (fig. S4, C and D). In addition, a VHH-based chimera in which VHH(GFP) was fused to PSMD4 showed robust TPD activity toward GFP-tagged ODC (Fig. 4, F and G). Next, we set out to target BRD4, a protein whose inhibition or depletion has been shown to be therapeutically relevant as an anticancer intervention. The ADRM1-scFv(BRD4) chimera substantially lowered endogenous BRD4 protein levels without affecting *BRD4* mRNA expression, accelerating BRD4 turnover as well (Fig. 4, H and I, and fig. S4F). Furthermore, the system displayed high substrate specificity from IB analyses: scFv(c-Fos) degraders did not affect the protein levels of <sup>Flag</sup>TDP43, <sup>HA</sup>Tau, or <sup>GFP</sup>ODC, whereas scFv(Flag) or scFv(HA) degraders did not alter c-Fos levels (Fig. 4, J and K, figs. S2, G and H, and S4E).

To evaluate the global impact of Protea-Tac degraders, we performed a quantitative proteomic study using TMT-based quantitative proteomics. Protein samples from five independent cultures of A549 cells overexpressing c-Fos transfected with scFv(c-Fos)-ADRM1 or individual modules were labeled with isobaric TMT reagents and analyzed by liquid chromatography–tandem mass spectrometry (LC-MS/MS), identifying a total of 6295 proteins (Fig. 4L). The principal components analysis showed separation between the control and Protea-Tac groups (fig. S4H). Applying fold change (FC) thresholds of >1.2 or <0.83 and a *q* value of <0.05, only c-Fos and ADRM1 were significantly reduced, whereas the broader proteome remained largely unchanged (Fig. 4M). The pronounced reduction in c-Fos protein levels, as confirmed by both biochemical and TMT-MS assays, underscores high selectivity of the Protea-Tac system, driven by tAb specificity. The observed reduction in ADRM1 abundance likely reflects the faster turnover of the scFv(c-Fos)-ADRM1 fusion compared with ADRM1 expressed separately with scFv(c-Fos). In addition, the absence of significant changes in c-Fos downstream proteins likely reflects the artificial c-Fos-overexpression context used for TMT-MS/MS to ensure reliable quantification of on-target c-Fos depletion (fig. S4G). Even after stable overexpression of Protea-Tac chimeras, we detected no marked changes in native proteasome profiles, in-gel proteasome activity, proteasome abundance, or selected markers of the UPS and related pathways (Fig. 4, N and O, and fig. S4, I to K). Together, these results indicate that modifying proteasomal UbRs with tAbs enables selective target degradation without markedly perturbing 26S proteasome function or the broader proteome under the conditions tested.

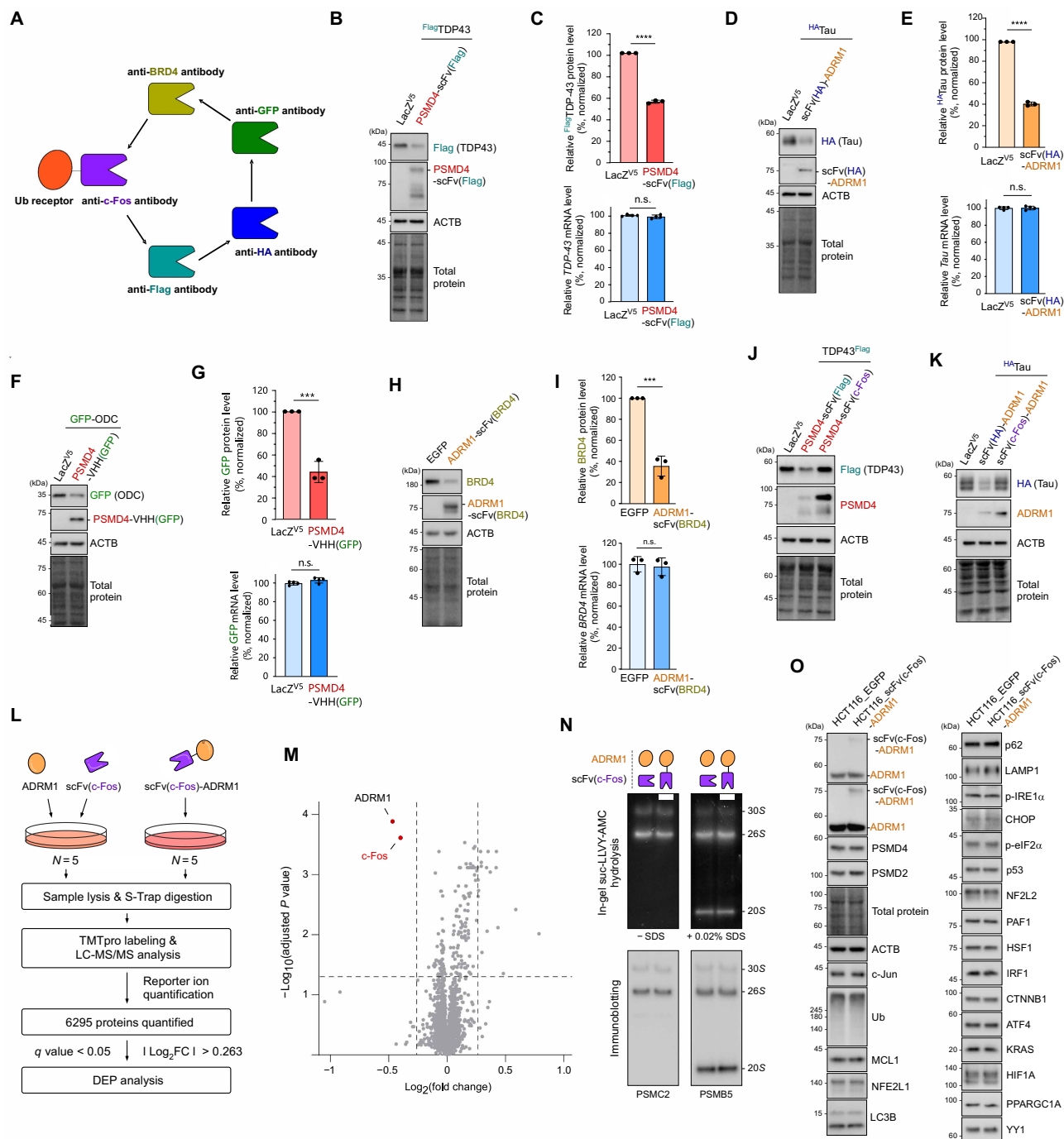
## In vivo target degradation and antitumor efficacy of Protea-Tac

To assess whether Protea-Tac-mediated c-Fos depletion suppresses tumor growth in vivo, we generated xenografts using stable MCF7 and HCT116 cell lines overexpressing scFv(c-Fos)-ADRM1 and their corresponding EGFP-expressing controls (Fig. 1J and fig. S1N), and subcutaneously injected equivalent quantities of these stable cells into 6-week-old male nonobese diabetic (NOD)/severe combined immunodeficient (SCID)/interleukin-2γ (IL-2γ)-receptor-null (NSG) mice. Tumor volumes and body weights were monitored every 3 days for 5 to 7 weeks. At the experimental endpoint (day 32 or day 50), mice bearing Protea-Tac-expressing tumors showed substantially delayed tumor growth compared with controls [ $978.1 \pm 285.5 \text{ mm}^3$  in MCF7\_EGFP tumors versus  $92.4 \pm 18.9 \text{ mm}^3$  in MCF7\_scFv(c-Fos)-ADRM1 tumors;  $1244.8 \pm 459.8 \text{ mm}^3$  in HCT116\_EGFP tumors versus  $176.3 \pm 44.8 \text{ mm}^3$  in HCT116\_scFv(c-Fos)-ADRM1 tumors] (Fig. 5, A and B). In stark contrast, there were no significant differences in body weights between the groups with MCF7 and HCT116 tumors (Fig. 5C and fig. S5A).

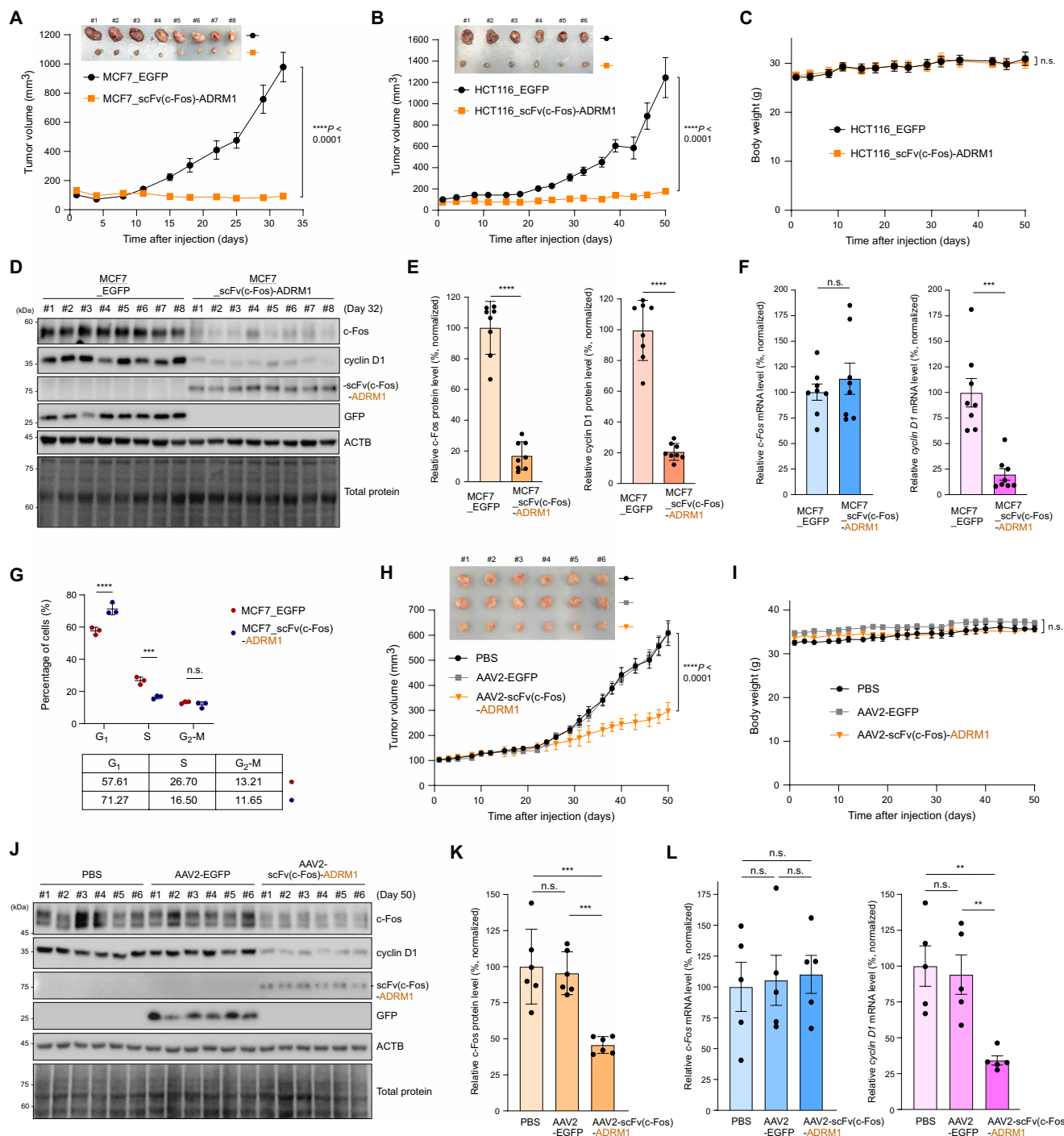
IB examination of dissected tumor lysates revealed marked depletion of c-Fos protein in scFv(c-Fos)-ADRM1-expressing tumors but not in EGFP-expressing controls (Fig. 5D and fig. S5B), demonstrating in vivo TPD activity of the Protea-Tac system. Furthermore, cyclin D1 was also significantly reduced, but there was little change in PARP cleavage (an apoptotic marker) and ERK1/2 and p38 phosphorylation (c-Fos upstream markers) (Fig. 5E and fig. S5, C to E). Analysis of mRNA levels from the tumor tissues revealed a significant down-regulation of *cyclin D1* expression, a marked up-regulation of Protea-Tac mRNA expression, and no change in endogenous *c-Fos* mRNA levels (Fig. 5F and fig. S5, F and G). Flow cytometry of stable cell lines used in xenograft assays further confirmed a pronounced delay in cell cycle progression, particularly at the G<sub>1</sub>-S transition, in tumors expressing the c-Fos-targeting Protea-Tac degraders (Fig. 5G and fig. S5H). These findings demonstrate that Protea-Tac degraders have substantial antitumor actions in vivo, which are mediated predominantly through cell cycle arrest rather than apoptosis activation.

To evaluate the therapeutic potential of Protea-Tac, we used AAV-based expression and MCF7 xenografts in NSG mice. Intratumoral injection with AAV2 encoding either EGFP or scFv(c-Fos)-ADRM1 revealed that mice receiving Protea-Tac exhibited a significant delay in tumor progression compared to both phosphate-buffered saline (PBS)- and AAV2-EGFP-injected controls ( $608.5 \pm 121.9 \text{ mm}^3$  for PBS versus  $610.2 \pm 92.4 \text{ mm}^3$  in AAV2-EGFP versus  $296.9 \pm 34.1 \text{ mm}^3$  for AAV2-scFv(c-Fos)-ADRM1; \*\*\*\**P* < 0.0001) (Fig. 5H). Body weights remained comparable across all groups, with no evident treatment-associated weight loss (Fig. 5I). IB analyses of tumor lysates mirrored the molecular profiles observed in stable cell-derived xenografts, showing robust expression of AAV2-scFv(c-Fos)-ADRM1, efficient degradation of c-Fos, and a marked reduction in cyclin D1 protein levels (Fig. 5, J and K, and fig. S5I). Unlike *c-Fos* mRNA, which was unchanged in the AAV2-Protea-Tac group, *cyclin D1* mRNA expression was markedly reduced (Fig. 5L).

To further evaluate AAV2-mediated Protea-Tac delivery in an immunocompetent setting, we established a syngeneic mouse tumor model by subcutaneously implanting CT26 cells into BALB/c mice. When tumors reached  $\sim 150 \text{ mm}^3$ , mice were intratumorally administered PBS or AAV2 encoding EGFP or scFv(c-Fos)-ADRM1



**Fig. 4. Protea-Tac-mediated degradation is highly selective and modular via UbR and tAb exchanges.** (A) Schematic of Protea-Tac modularity through interchangeable tAb units. (B to I) A549 cells transiently expressing Flag-TDP43, HA-Tau, GFP-ODC, or endogenous BRD4 were treated with cognate Protea-Tac degraders or controls and analyzed by SDS-PAGE/IB, total protein staining, and qRT-PCR. Protein quantification was normalized to total protein unless indicated otherwise; qRT-PCR values were normalized to GAPDH. (B and C) PSMD4-scFv(Flag) degraded Flag-TDP43 without reducing TDP43 mRNA. Protein:  $N = 3$ , means  $\pm$  SD, \*\*\*\*  $P < 0.0001$ ; mRNA:  $N = 4$ , n.s.; two-tailed Student's  $t$  test. (D and E) scFv(HA)-ADRM1 degraded HA-Tau without reducing tau mRNA. Protein:  $N = 3$ , means  $\pm$  SD, \*\*\*\*  $P < 0.0001$ ; mRNA:  $N = 4$ , n.s.; two-tailed Student's  $t$  test. (F and G) PSMD4-VHH(GFP) degraded GFP-tagged ODC without reducing GFP mRNA. Protein:  $N = 3$ , means  $\pm$  SD, \*\*\*  $P < 0.001$ ; mRNA:  $N = 4$ , n.s.; two-tailed Student's  $t$  test. (H and I) Lentiviral ADRM1-scFv(BRD4) reduced endogenous BRD4 protein but not BRD4 mRNA. Protein and mRNA:  $N = 3$ , means  $\pm$  SD, \*\*\*  $P < 0.001$  or ns; two-tailed Student's  $t$  test. (J and K) Mismatched c-Fos-targeting Protea-Tac did not degrade Flag-TDP43 or HA-Tau. (L) TMT-LC-MS/MS global proteome profiling compared A549 stably overexpressing c-Fos cells transfected by scFv(c-Fos)-ADRM1 or ADRM1 plus scFv(c-Fos) controls. Among 6295 quantified proteins, 16 differentially expressed proteins (DEPs) were identified using  $|\log_2 FC| > 0.263$  and adjusted  $P$  value  $< 0.05$ . (M) Volcano plot of differentially expressed proteins, with c-Fos and ADRM1 highlighted in red. (N) Native PAGE analysis of 26S proteasomes by in-gel suc-LLVY-AMC hydrolysis and IB; 0.02% SDS activated 20S proteasomes. (O) Endogenous protein levels in HCT116\_EGFP and HCT116\_scFv(c-Fos)-ADRM1 stable cells were analyzed by IB. See fig. S4.



**Fig. 5. Protea-Tac-induced c-Fos degradation leads to antitumor efficacy in vivo.** (A) MCF7 cells stably expressing EGFP or scFv(c-Fos)-ADRM1 were subcutaneously inoculated into 6-week-old male NSG mice, and tumor volumes were measured twice weekly for ~5 weeks. Data are means  $\pm$  SEM,  $N = 8$  per group, \*\*\*\* $P < 0.0001$ ; Mann-Whitney  $U$  test. Insets show tumors harvested on day 32. (B) Tumors harvested on day 50 from HCT116\_EGFP or HCT116\_scFv(c-Fos)-ADRM1 xenografts established as in (A). (C) Body weight changes during the HCT116 xenograft experiment. (D) IB analysis of tumor lysates. (E) Quantification of c-Fos and cyclin D1 proteins, normalized to total protein signals. Data are means  $\pm$  SD,  $N = 8$ , \*\*\*\* $P < 0.0001$ ; two-tailed Student's  $t$  test. (F) qRT-PCR analysis of *c-Fos* and *cyclin D1* mRNAs, normalized to *GAPDH*. Data are means  $\pm$  SEM,  $N = 8$ , \*\*\* $P < 0.001$ , n.s.; two-tailed Student's  $t$  test. (G) Cell cycle distribution of MCF7 stable cells used for xenograft assays, analyzed by DAPI staining and flow cytometry.  $N = 3$ , \*\*\* $P < 0.001$ , \*\*\*\* $P < 0.0001$ , n.s.; two-way ANOVA with Sidak's post hoc test. (H) MCF7-xenografted NSG mice were intratumorally injected three times per week with PBS, AAV2-EGFP, or AAV2-scFv(c-Fos)-ADRM1, and tumor growth was measured three times per week for 50 days. Data are means  $\pm$  SEM,  $N = 6$  per group, \*\*\*\* $P < 0.0001$ ; two-way ANOVA with Sidak's post hoc test. (I) Body weight changes during the AAV2-delivery experiment. (J) IB analysis of tumor lysates. (K) c-Fos protein quantification. Data are means  $\pm$  SD,  $N = 6$ , \*\*\* $P < 0.001$ , n.s.; one-way ANOVA with Tukey's post hoc test. (L) qRT-PCR analysis of *c-Fos* and *cyclin D1* mRNAs. Data are means  $\pm$  SEM,  $N = 5$ , \*\* $P < 0.01$ ; one-way ANOVA with Tukey's post hoc test. See figs. S5 and S6.

once weekly for a total of three injections. AAV2-mediated expression of scFv(c-Fos)-ADRM1 significantly suppressed tumor growth compared with the PBS- and AAV2-EGFP-treated groups (fig. S6, A and B). IB analyses of tumor lysates showed reduced c-Fos and cyclin D1 protein levels in the AAV2-scFv(c-Fos)-ADRM1 group (fig. S6, D and E). Body weights remained comparable across the groups, and serum biochemical analyses revealed no significant differences in alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP), uric acid (UA), creatinine (CREA), or blood urea nitrogen (BUN) levels at the experimental endpoint (fig. S6, C, F, and G). Serum IL-6 levels were also not significantly elevated at the endpoint (fig. S6H).

### Nonviral mRNA-LNP delivery of Protea-Tac induces target degradation and suppresses tumor growth in vivo

To further explore delivery strategies beyond AAV-based expression, we next evaluated whether Protea-Tac could be delivered as mRNA formulated in LNPs. Although ADRM1-based Protea-Tacs were used in our preceding in vivo studies, PSMD4-based Protea-Tac also exhibited robust degradation activity and was therefore selected for evaluation in the context of mRNA delivery. Consistent with our earlier findings, stable expression of PSMD4-scFv(c-Fos), but not scFv(c-Fos) alone, reduced endogenous c-Fos protein levels in HCT116 cells without altering c-Fos mRNA levels. This reduction was accompanied by decreased cyclin D1 protein and mRNA levels (fig. S7, A and B). Similarly, transfection of A549 cells with mRNA encoding PSMD4-scFv(c-Fos) reduced endogenous c-Fos protein levels while leaving c-Fos mRNA levels unchanged. Cyclin D1 expression was also reduced under these conditions (fig. S7, C to E).

We next evaluated mRNA-LNP-mediated Protea-Tac delivery in an A549 xenograft model. Mice bearing A549 tumors received subcutaneous injections of PBS or LNP formulations containing mRNA encoding EGFP, scFv(c-Fos), or PSMD4-scFv(c-Fos) at the indicated time points. Mice treated with LNP formulations containing mRNA encoding PSMD4-scFv(c-Fos) exhibited significantly delayed tumor growth compared with mice receiving PBS or LNP formulations containing mRNA encoding EGFP or scFv(c-Fos) (Fig. 6, A and B). IB analyses of tumor lysates confirmed expression of the encoded proteins and revealed marked reductions in c-Fos and cyclin D1 protein levels specifically in tumors treated with PSMD4-scFv(c-Fos) encoding mRNA-LNPs, without evident PARP cleavage (Fig. 6, C and D). Analysis of tumor mRNA levels showed that c-Fos mRNA remained unchanged, whereas cyclin D1 mRNA was significantly reduced (Fig. 6E). These results are consistent with post-translational depletion of c-Fos and the consequent suppression of cyclin D1 expression. Serum biochemical analyses revealed no significant differences in ALT, AST, BUN, or CREA levels among all groups (fig. S7F), indicating no detectable changes in hepatic or renal injury markers under the conditions tested. Together with our AAV-mediated experiments, these findings provide proof of concept that Protea-Tac can be delivered through both viral and nonviral modalities to show antitumor efficacy in vivo.

### DISCUSSION

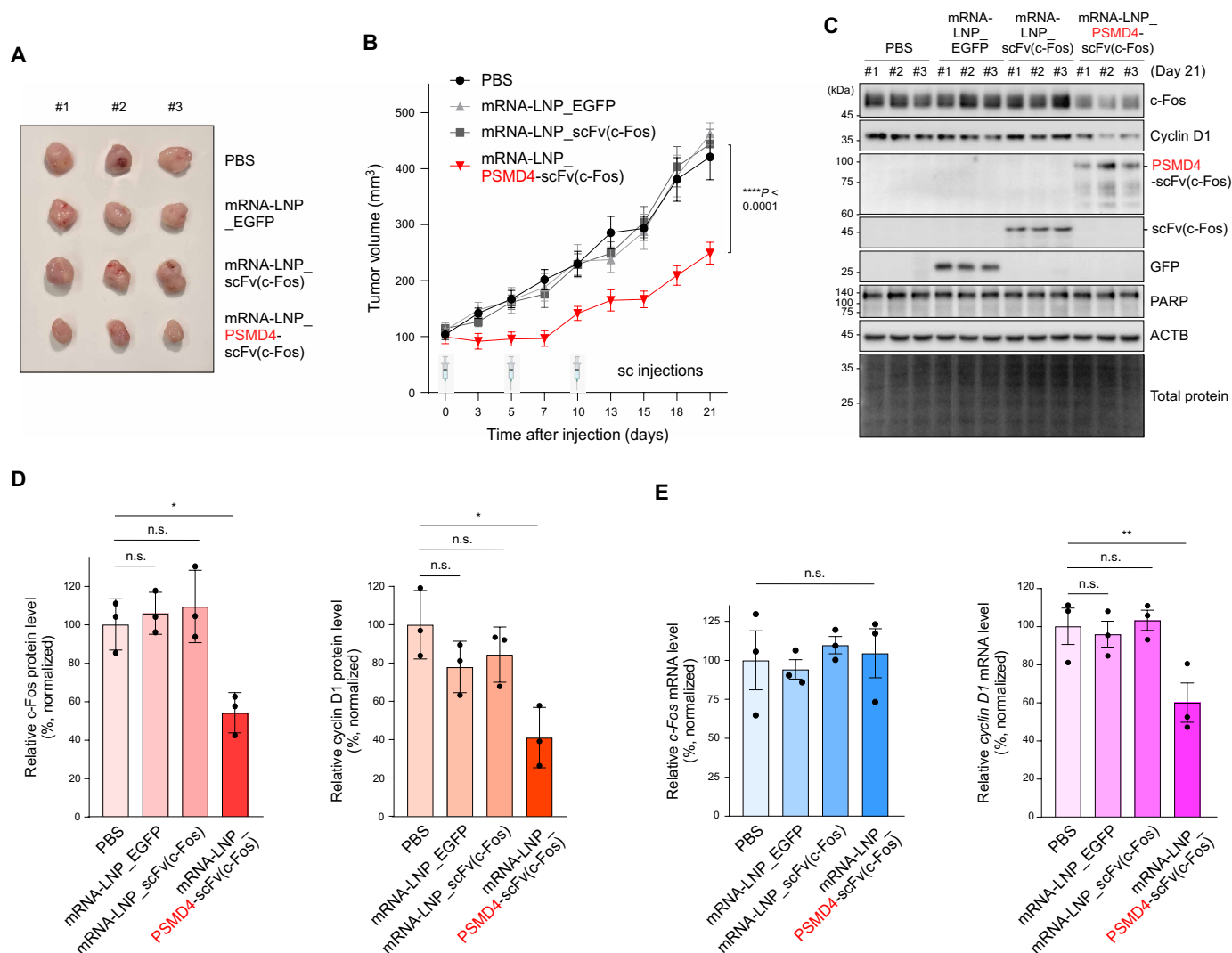
The 26S proteasome is the sole adenosine triphosphate (ATP)-dependent protease in the eukaryotic cytoplasm and nucleus, which uses Ub chains for the recognition of to-be-degraded substrates. Ub chains are primarily used to localize substrates to the 26S proteasome. Unless

a substrate lacks a flexible, unstructured region (the “initiation site”) that triggers unfolding and translocation via the AAA+ PSMC pore loop, this tethering alone is usually sufficient to cause spontaneous substrate hydrolysis in the 20S core particle (59–61). On the basis of this mechanism, we developed Protea-Tac by fusing a proteasomal UbR to an intracellular tAb, thereby replacing Ub chain-mediated substrate recruitment with direct antibody-guided engagement of the 26S proteasome. This modular architecture enabled degradation of overexpressed c-Fos and further demonstrated its generalizability through tag-directed chimeras against TDP43, tau, and ODC. Moreover, its TPD activity against endogenous c-Fos and BRD4 proteins indicates that Protea-Tac is not restricted to short-lived substrates but also active toward comparatively stable endogenous proteins. By promoting Ub-independent proteasomal degradation, Protea-Tac bypasses the substrate ubiquitination and E3 ligase recruitment steps required by conventional ubiquitination-inducing degraders, including PROTACs and many MGDs. This feature may be advantageous in settings where the engaged Ub transfer machinery is compromised, a recognized mechanism of resistance to conventional degraders (62, 63).

Recent direct-to-proteasome small-molecule degraders have established that chemically induced recruitment of neo-substrates to UbRs or other RP components can induce Ub-independent TPD (46–50). These approaches offer important advantages inherent to small-molecule modalities, including cell permeability and opportunities for pharmacological optimization. Protea-Tac should therefore be viewed as a complementary protein-based strategy rather than a competing replacement. Its distinction lies not simply in its molecular format, but in the source of target recognition: Whereas chemical degraders depend on the availability of a suitable small-molecule ligand for each neo-substrate, Protea-Tac encodes target recognition in a modular intracellular binder. This does not eliminate the need for optimization but shifts an important early bottleneck from small-molecule warhead discovery toward the identification and engineering of binders with suitable specificity and geometry. Such a framework may be particularly useful for ligand-poor proteins, posttranslationally modified disease-associated proteins (e.g., phosphorylated effectors) and for applications requiring conformation- or modification-selective target engagement (64–67).

Beyond expanding the range of accessible targets, the proteasome-centered design of Protea-Tac may offer additional practical advantages. Protea-Tac may benefit from the substantial cellular abundance of 26S proteasomes as robust target degradation was observed despite the apparent remodeling of only a minor subpopulation of cellular proteasomes (23, 24). This limited extent of remodeling may help preserve global proteasome activity and minimize nonspecific perturbation of endogenous protein turnover, whereas precise antibody-guided recognition contributes to the observed target selectivity. Our affinity-tuning analyses further suggest that maximal substrate-binding affinity is not necessarily required for efficient Protea-Tac activity; rather, a dynamic interaction with the recruited substrate may favor productive turnover in permissive configurations. Together with ongoing advances in antibody and de novo mini-binder engineering, the modularity of this platform may facilitate the development of a broader repertoire of Protea-Tac degraders for disease-associated proteins that remain difficult to engage with small-molecule ligands (68).

The modularity of Protea-Tac does not imply that every proteasome-tethering configuration will be equally productive. Protea-Tac activity was also dependent on the proteasomal subunit used to construct



**Fig. 6. mRNA-LNP-mediated Protea-Tac expression inhibits tumor growth in vivo.** (A) A549-xenografted 6-week-old female BALB/cAnN-Foxn1<sup>nu</sup>/CrI mice received subcutaneous (sc) injections of PBS or LNP formulations containing mRNA encoding EGFP, scFv(c-Fos), or PSMD4-scFv(c-Fos). Representative image of tumors harvested is shown. (B) Tumor volumes were measured twice a week for 21 days. LNP formulations were administered subcutaneously every 5 days for a total of three doses. Each formulation contained 40 µg of mRNA per dose. Data are presented as means ± SEM (*N* = 6 per group); two-way ANOVA followed by Sidak's post hoc test (\*\*\*\**P* < 0.0001). (C) IB analysis of whole-tumor lysates with indicated antibodies. (D) Quantification of c-Fos (left) and cyclin D1 (right) protein levels, normalized to total protein signals. Bars represent the means ± SD (*N* = 3). n.s., not significant, \**P* < 0.05 (one-way ANOVA with Tukey's post hoc test). (E) As in (D), but mRNA levels of c-Fos and cyclin D1 were determined using qRT-PCR and normalized to GAPDH. Bars represent the means ± SEM (*N* = 3). n.s., not significant, \*\**P* < 0.01 (one-way ANOVA with Tukey's post hoc test).

the chimera. Whereas PSMD4- and ADRM1-based Protea-Tacs efficiently reduced c-Fos, chimeras constructed with PSMD2, PSMA4, PSMB2, or PSMC5 did not induce appreciable target degradation. In addition, extending the flexible linker did not restore TPD activity in the inactive configurations tested. Although the structural basis for these differences remains to be determined, these findings are consistent with the concept that recruitment to the proteasome is necessary but not invariably sufficient for productive degradation (46). Proteasomal degradation requires not only substrate tethering but also a global conformational transition of the RP subcomplex initiated by substrate engagement with the AAA+ ATPase pore (69, 70). Because the location and accessibility of degradation-initiation regions vary across different neo-substrates, further development

may benefit from structure-guided, substrate-specific optimization of the targeting binder, linker architecture, and selected proteasomal subunit.

Despite inherent challenges related to intracellular delivery and expression, Protea-Tac represents a distinct and promising protein-based TPD strategy. Our in vivo studies demonstrate that this platform is compatible with both viral and nonviral delivery formats. First, using AAV2-mediated intratumoral delivery, we achieved robust in vivo c-Fos degradation and antitumor efficacy not only in MCF7 xenografts but also in an immunocompetent CT26 syngeneic mouse tumor model. Given that multiple recombinant AAV-based products have already received clinical regulatory approval, this delivery route holds translational potential (71). In parallel, we established

proof of concept for a nonviral approach: subcutaneous administration of mRNA-LNPs encoding PSMD4-scFv(c-Fos) successfully degraded c-Fos in vivo and attenuated tumor growth in A549 xenografts (72). Meanwhile, the immunogenicity of the delivery vehicles and the Protea-Tac protein itself will also require dedicated investigation, including innate and adaptive responses (73, 74). Thus, the present study does not establish a clinically optimized therapeutic modality; rather, it defines a modular framework for direct, antibody-guided, Ub-independent degradation through engineered 26S proteasomes and provides in vivo proof of concept.

## MATERIALS AND METHODS

### Study design

This study used cell-based assays and mouse tumor models to evaluate the target-degrading activity, mechanism of action, and in vivo antitumor effects of Protea-Tac. Sample sizes for cell-based and animal experiments were selected on the basis of commonly used group sizes in comparable biochemical and preclinical tumor studies and were not determined by formal power calculations. For animal studies, mice were allocated to treatment groups after tumor establishment by balancing tumor sizes across groups before injection. Investigators were not blinded to treatment allocation during treatment administration, tumor measurement, or endpoint analyses. Tumor volume, body weight, target protein depletion, and endpoint biochemical parameters were used as prespecified endpoints. Data were excluded only when samples failed technical quality control criteria, such as unsuccessful tumor establishment, sample loss during processing, or insufficient protein or RNA quality for the planned analysis. All animal experiments were performed in accordance with protocols approved by the relevant Institutional Animal Care and Use Committees (IACUCs) and with institutional guidelines for ethical and humane animal care. The study followed the principles of the 3Rs by prioritizing cell-based experiments before in vivo studies, using the minimum number of animals needed to evaluate tumor-growth and molecular endpoints, and monitoring animals regularly to minimize pain or distress. Details regarding the IACUC, animal strain, sex, age, vendor, housing, tumor implantation, treatment route, dosing schedule, and tissue collection are provided below.

### Antibodies and reagents

The following primary antibodies were used: anti-c-Fos [Cell Signaling Technology (CST), 1:2000 for IB], anti-PSMD4 [Thermo Fisher Scientific (TFS), 1:1000], anti-ADRM1 (TFS, 1:1000), anti-PSMA4 (Enzo Life Sciences, 1:1000), anti-V5 (TFS, 1:3000), anti-ACTB (MilliporeSigma, 1:3000), anti-GAPDH [Santa Cruz Biotechnology (SCBT), 1:3000], anti-MCL1 (CST, 1:2000), Streptavidin-horseradish peroxidase (HRP) (MilliporeSigma, 1:1000), anti-cyclin D1 (CST, 1:2000), anti-His (CST, 1:1000), anti-VCP (TFS, 1:3000), anti-Flag (TFS, 1:1000), anti-HA (MilliporeSigma, 1:3000), anti-GFP (SCBT, 1:3000), anti-PSMC2 (SCBT, 1:1000), anti-PSMB5 (TFS, 1:1000), anti-PSMB6 (TFS, 1:1000), anti-PSMA3 (Enzo Life Sciences, 1:1000), anti-PSMD1 (SCBT, 1:1000), anti-LC3B (MilliporeSigma, 1:1000), anti-Ub (SCBT, 1:3000), anti-NFE2L1 (TFS, 1:3000), anti-p62 (Abcam, 1:3000), anti-LAMP1 (SCBT, 1:1000), anti-p-IRE1 $\alpha$  (Ser<sup>724</sup>, TFS, 1:1000), anti-CHOP (TFS, 1:1000), anti-p-eIF2 $\alpha$  (Abclonal, 1:1000), anti-c-Jun (CST, 1:2000), anti-p53 (TFS, 1:1000), anti-PAF1 (Abnova, 1:1000), anti-HSF1 (CST, 1:1000), anti-IRF-1 (CST, 1:1000), anti-CTNNB1 (CST, 1:1000), anti-ATF4 (TFS, 1:1000), anti-RAS (CST,

1:1000), anti-HIF-1 (Novus, 1:1000), anti-YY1 (Abclonal, 1:1000), anti-NFE2L2 (abcam, 1:1000), anti-PPARGC1A (Abclonal, 1:1000), anti-PARP (TFS, 1:1000), anti-ERK1/2 (CST, 1:1000), anti-p-ERK1/2 (CST, 1:1000), anti-p38 (CST, 1:1000), and anti-p-p38 (CST, 1:1000). Secondary antibodies included HRP-conjugated anti-mouse IgG, anti-rabbit IgG, and anti-rat IgG (all from TFS, used at 1:10,000 for IB). Sources of major biochemical reagents are as follows: MG132 (AG Scientific), MLN7243 (Cayman Chemical), Bafilomycin A1 (Cayman Chemical), suc-LLVY-AMC (Bachem), and No-Stain Protein Labeling Reagent (TFS). All scFvs and VHs were derived from plasmids deposited at Addgene (182090, 198303, 198302, and 166518).

### Mammalian cell culture and transient expression

Mammalian cell lines used in this study, including A549 [Korean Cell Line Bank (KCLB) no.: 10185], HCT116 (KCLB no.: 10247), MCF7 (KCLB no.: 30022), and their derivatives stably expressing EGFP and scFv(c-Fos)-ADRM1, were cultured in RPMI (Wellgene), whereas HEK293T cells were cultured in Dulbecco's modified Eagle's medium (DMEM) (Wellgene), supplemented with 10% heat-inactivated fetal bovine serum (Gibco), penicillin (100 U/ml), streptomycin (100  $\mu$ g/ml), and 2 mM L-glutamine. Cells were maintained at 37°C in a humidified atmosphere containing 5% CO<sub>2</sub>. For transient overexpression, cells were seeded to reach ~70% confluence and transfected with indicated plasmid DNA using Lipofectamine 3000 (TFS) according to the manufacturer's instructions. The total amount of plasmid DNA was kept constant within each experimental set (2 or 4  $\mu$ g of total plasmid DNA in a 6-well plate). After incubation, cells were washed with PBS and lysed in RIPA buffer [50 mM tris-HCl (pH 8.0), 1% NP-40, 0.5% deoxycholate, 0.1% SDS, and 150 mM NaCl] supplemented with protease inhibitor cocktails or other indicated buffers. Whole-cell lysates (WCLs) were clarified by centrifugation at 13,000 rpm for 15 min at 4°C, mixed with 2 $\times$  SDS sample buffer, and denatured at 85°C for 15 min before SDS-polyacrylamide gel electrophoresis (PAGE) and subsequent analyses. For chase analysis, cells were treated with cycloheximide (80  $\mu$ g/ml) at time zero, and samples were collected at indicated chase points.

### RNA isolation and quantitative RT-PCR

Total RNA from cultured cells was extracted using TRIzol (Favorgen), followed by additional purification with RNeasy Mini Columns (Qiagen) with on-column DNase I treatment. cDNA synthesis was performed using a reverse transcription polymerase chain reaction (RT-PCR) premix (Bioneer), and quantitative RT-PCR (qRT-PCR) was conducted using 1:20-diluted cDNA, SYBR qPCR Master Mix (Bioneer), and 10  $\mu$ M primers. The target gene-specific primer sequences were as follows: for *GAPDH*, forward 5'-AGGGCCCTGACAACCTCTTTT-3' and reverse 5'-AGGGGTCTACATGGCAACTG-3'; for *c-Fos*, forward 5'-CACTCCAAGCGGAGACAGAC-3' and reverse 5'-AG-GTCATCAGGGATCTTGCAG-3'; for *cyclin D1*, forward 5'-GCT-GCGAAGTGGAACCATC-3' and reverse 5'-CCTCCTTCTGCACACATTGTGA-3'; for *c-Fos (K0)*, forward 5'-TACCCCTACGACGTGCCCCG-3' and reverse 5'-GAGTCTGCGGGTGAAGTGGTAG-3'; for *BRD4*, forward 5'-ACCTCCAACCTAACAGGCC-3' and reverse 5'-TTTC-CATAGTGTCTTGAGCACC-3'; for *TDP43*, forward 5'-GGGTAACCGAAGATGAGAACG-3' and reverse 5'-CTGGGCTGTAAACCGTGGAG-3'; for *tau*, forward 5'-CCAAGTGTGGCTCATTAGGCA-3' and reverse 5'-CCAATCTTCGACTGGACTCTGT-3'; for

*GFP*, forward 5'-AAGCAGAAGAACGGCATCAA-3' and reverse 5'-GGGGGTGTCTGCTGGTAGT-3'; and for scFv(c-Fos)-ADRM1, forward 5'-CCGAAGATCTGGGAGTGTACT-3' and reverse 5'-CACC-AAGTACTTGTGGAGGC-3', where the scFv primers were designed to specifically amplify the scFv region, avoiding endogenous *ADRM1*. Gene expression levels were normalized to *GAPDH*, and statistical significance was evaluated using a two-tailed Student's *t* test with *P* < 0.05 considered significant.

### Soluble/insoluble fractionation

Cells expressing Protea-Tac degraders were washed with ice-cold PBS and lysed in RIPA buffer supplemented with protease inhibitor cocktails. WCLs were centrifuged at 16,000g for 30 min at 4°C to separate supernatants (RIPA-soluble fractions), which were boiled at 85°C for 15 min in 2× SDS sample buffer. Pellets (RIPA-insoluble fractions) were washed with lysis buffer, resuspended in SDS sample buffer, and boiled at 100°C for 15 min.

### Affinity purification and native PAGE analysis of 26S proteasomes

For affinity purification using biotin tags, cells were harvested in lysis buffer [25 mM Tris-HCl (pH 7.5), 10% glycerol, 5 mM MgCl<sub>2</sub>, 1 mM ATP, 1 mM dithiothreitol (DTT), and protease inhibitor cocktails] and homogenized with a Dounce homogenizer. The lysates were incubated with streptavidin-conjugated agarose beads overnight at 4°C. The beads were washed five times with wash buffer [20 mM Tris (pH 7.5), 15% glycerol, 1 mM EDTA, 150 mM KCl, 0.05% NP-40, 1 mM DTT, and 0.2 mM phenylmethylsulfonyl fluoride]. Next, the precipitated proteins were eluted from the resin by incubating in TEV cleavage buffer [50 mM Tris-HCl (pH 7.5), 1 mM ATP, 10% glycerol, and TEV protease] for 3 hours at 30°C. Then, the eluted proteins were concentrated with Amicon Ultra-0.5 centrifugal filter units (MilliporeSigma). Proteins were resuspended in Tris-Glycine Native Sample Buffer (TFS) and resolved on 3 to 8% NuPAGE Tris-Acetate native gels (TFS) at 150 V for 4 to 5 hours. Gels were incubated in activity assay buffer (20 mM Tris, 1 mM ATP, and 5 mM MgCl<sub>2</sub>) containing 100 μM suc-LLVY-AMC to visualize proteasome complexes. To activate 20S proteasomes in native gels, 0.02% SDS was added. After fluorescence-based in-gel activity assay, separated samples in the gel were transferred to polyvinylidene difluoride membranes for IB analysis.

### Flow cytometry for c-Fos degradation and cell cycle distribution

EGFP-c-Fos stable cells were transfected with either tricistronic scFv(c-Fos)-T2A-ADRM1-P2A-mCherry or bicistronic scFv(c-Fos)-ADRM1-P2A-mCherry plasmids and incubated for 72 hours. Cells were harvested, washed, and resuspended in FACS buffer (0.2% bovine serum albumin (BSA), 0.1% sodium azide, and 2 mM EDTA in PBS). Flow cytometry was performed on a BD LSRFortessa X-20 using fluorescein isothiocyanate (FITC) and phycoerythrin (PE) channels. Data were analyzed using the FlowJo software (version 10.9.0). For DNA content analysis, EGFP or scFv(c-Fos)-ADRM1-expressing cells were fixed in 70% ethanol overnight, washed, and stained with 4',6-diamidino-2-phenylindole (DAPI)/Triton X-100 at room temperature (RT) for 30 min in the dark. Flow cytometry was performed using LSRFortessa (BD Bioscience) with ultraviolet excitation at 340 to 380 nm and analyzed using FlowJo to quantify the distribution across G<sub>0</sub>-G<sub>1</sub>, S, and G<sub>2</sub>-M phases.

### Affinity tuning of scFv(c-Fos) and affinity measurement

To change the binding affinity of scFv(c-Fos), random mutagenesis of six central residues within the CDR3 region was performed using primers containing NNK degenerate codons. The resulting mutant libraries were cloned into the pMopac12 vector and transformed into *Escherichia coli* Jude1 cells. Individual colonies (*N* = 596) were cultured in 96-deep-well plates and induced with 0.5 mM isopropyl-β-D-thiogalactopyranoside (IPTG) at 20°C for 20 hours. Secreted scFv(c-Fos) variants were purified from culture supernatants using Ni-NTA affinity chromatography. To express recombinant glutathione *S*-transferase (GST)-tagged c-Fos, *E. coli* BL21(DE3) cells harboring the expression plasmids were induced with 0.5 mM IPTG, followed by incubation at 30°C for 16 hours. Cells were lysed, and clarified lysates were applied to a GST affinity column, eluted with 20 mM reduced glutathione, and buffer exchanged into PBS using Amicon Ultra centrifugal filters (10-kDa MWCO). Protein purity and molecular weight were verified by SDS-PAGE and protein staining, and concentrations were determined by absorbance at 280 nm using a NanoDrop spectrophotometer.

To assess antibody affinity, purified scFvs (200 ng per well) were coated onto 96-well immunoplates and incubated overnight at 4°C. After blocking with 3% BSA in PBS, serial dilutions of GST-c-Fos (ranging from 5 μM to 0.32 nM) were added to the plates and incubated for 1 hour at RT. Plates were then washed with PBST and incubated with HRP-conjugated anti-GST antibodies for additional 1 hour at RT. TMB-ELISA (3,3',5,5'-tetramethylbenzidine-enzyme-linked immunosorbent assay) substrates (TFS) were then applied for 20 min, and the reaction was stopped by the addition of 2 M H<sub>2</sub>SO<sub>4</sub>. Absorbance at 450 nm was measured using an Infinite 200 PRO NanoQuant microplate reader. Dose-response curves were generated, and median effective concentration (EC<sub>50</sub>) values were calculated from normalized values using GraphPad Prism (version 10.4.1).

### TMT-MS analysis

Protein samples were prepared from A549 cells stably expressing c-Fos after transfection with either ADRM1 and scFv(c-Fos) (control) or scFv(c-Fos)-ADRM1 degraders (Protea-Tac) for 64 hours in biological quintuplicates (*N* = 5 per group). Cells were washed three times with ice-cold PBS, snap frozen in liquid nitrogen, lysed in freshly prepared MS lysis buffer [4% SDS and 0.1 M Tris-HCl (pH 8.0)], heated at 95°C for 20 min, and sonicated. Lysates were cleared by centrifugation at 15,000 rpm for 20 min at RT, and protein concentrations were determined using a BCA assay (TFS). An aliquot containing 40 μg of total protein per sample was precipitated in pre-chilled acetone at -20°C overnight and collected by centrifugation at 14,000 rpm for 10 min at 4°C. After air-drying, pellets were resuspended in denaturing buffer [5% SDS, 10 mM TCEP, 50 mM CAA, and 50 mM Hepes (pH 8.5)], reduced, alkylated, and then acidified with phosphoric acid to pH ≤ 1. Following additional centrifugation, the samples were processed with S-Trap micro columns (Protifi), washed with methanol-containing buffer, and digested on-column with trypsin (1:25 enzyme-to-protein ratio) at 47°C for 2 hours. Peptides were eluted sequentially using three buffers (50 mM Hepes, 0.2% formic acid in water, and 0.2% formic acid in 50% acetonitrile), pooled, and dried under vacuum. The resulting peptides were labeled using TMTpro 16plex reagents (TFS; lot #XL348283) according to the manufacturer's instructions. After labeling for 15 min and quenching with 5% hydroxylamine, labeled peptides were combined in equal amounts across all channels, desalted using Sep-Pak tC18

cartridges (Waters), and dried again by vacuum centrifugation. To reduce ratio compression and systematic bias, the TMT channels were systematically assigned rather than randomized: HEK293T reference (channel 126), controls (127N–131C), and Protea-Tac groups (132N–134N). Pooled peptides were fractionated under high-pH conditions using an Agilent 1260 high-performance liquid chromatography system equipped with a ZORBAX 300Extend-C18 column (4.6 mm by 150 mm, 3.5  $\mu$ m; Agilent). Peptides were separated using a 55-min gradient, increasing buffer B [15 mM ammonium hydroxide in 90% acetonitrile (pH 10.0)] from 5 to 35% over 43 min at a flow rate of 0.2 to 1.0 ml/min. Ninety-six fractions were collected and concatenated into 24 fractions, which were vacuum dried and stored at  $-80^{\circ}\text{C}$  until MS analysis.

For LC-MS/MS analysis, each fraction was reconstituted in 10  $\mu$ l of 0.1% formic acid/2% acetonitrile solution, and 2  $\mu$ l was injected onto a trap column (75- $\mu$ m inside diameter by 2 cm, 3- $\mu$ m Pep-Map100 C18 beads) before separation on an EASY-Spray analytical C18 column (75  $\mu$ m by 50 cm, 2- $\mu$ m particle size; TFS) using an Ultimate 3000 ultrahigh-performance liquid chromatography system coupled to a Q-Exactive HF-X mass spectrometer (TFS). A 130-min linear gradient ramping solvent B (0.1% formic acid in acetonitrile) from 8 to 60% was applied at a flow rate of 300 nl/min. MS1 scans were acquired in positive ion mode over a mass range of mass/charge ratio ( $m/z$ ) 350 to 1800 at a resolution of 120,000 (full width at half maximum at  $m/z$  200), with an AGC target of  $3 \times 10^6$  and maximum injection time of 25 ms. The top 15 most intense precursor ions were selected for higher-energy collisional dissociation fragmentation using a 0.7-Th isolation window, normalized collision energy of 32%, and MS2 scans acquired at 30,000 resolution, with an AGC target of  $1 \times 10^5$  and maximum injection time of 80 ms. Raw MS data were analyzed using Proteome Discover version 3.1 (TFS), using the Sequest HT search engine and Percolator algorithm to control the false discovery rate (FDR) at 1% at both the peptide and protein levels. The MS/MS spectra were searched against the UniProt Swiss-Prot human canonical protein sequence database (retrieved on March 2024; 20,360 entries), appended with common contaminants. Precursor mass tolerance was set at  $\pm 10$  parts per million, and fragment mass tolerance at  $\pm 0.02$  Da. Peptide-spectrum matches were filtered to include only those with Sequest HT Xcorr  $\geq 1$ , high Percolator confidence, and reporter ion signal-to-noise ratios  $> 10$ , with spectra showing isolation interference  $> 70\%$  excluded. Quantification was performed using both unique and razor peptides, applying isotopic impurity correction factors provided by the manufacturer, and proteins identified with fewer than three unique peptides were excluded from further quantitative analysis. The TMT-MS proteomics data have been deposited in the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifier PXD063133.

### Tumor xenograft/allograft mouse model

All experiments related to animals were performed according to the protocols accredited by the IACUC of Seoul National University (IACUC no. SNU-240510-3-2), the Catholic University of Korea (IACUC no. CUK-IACUC-2026-015), and SML Biopharm (IACUC no. SML-BP-IACUC-2025-11). Mice were housed in a pathogen-free facility fully approved by the IACUC. EGFP or scFv(c-Fos)-ADRM1 stably expressing cells were suspended in a 1:1 mixture of PBS and Matrigel (Corning, USA), and 100  $\mu$ l of the suspension ( $5 \times 10^6$  cells) was subcutaneously injected into the right flank region of a 6-week-old

immunodeficient NOD.Cg-*Prkdc*<sup>scid</sup>*Il2rg*<sup>tm1Wjl</sup>/SzJ (NSG) male mice (the Jackson Laboratory) using a 26-gauge syringe. In case of CT26 allograft, CT26 cells were resuspended in PBS and 150  $\mu$ l ( $5 \times 10^6$  cells) of the suspension was subcutaneously injected into the right flank of 5-week-old female BALB/cAnNCrI (Orient Bio, Seongnam, Korea) using a 26-gauge syringe. To establish the A549 xenograft model, A549 cells were resuspended in a 1:1 mixture of PBS and Matrigel. A total volume of 100  $\mu$ l ( $3 \times 10^6$  cells) was injected subcutaneously into the right flank of 6-week-old female BALB/cAnN-*Foxn1*<sup>nu</sup>/CrI mice (Orient Bio, Seongnam, Korea) using a 26-gauge syringe. Tumor size and body weight were monitored twice a week using a caliper. Tumor volume was calculated using the formula: (length  $\times$  width<sup>2</sup>)/2. When the tumor of EGFP-expressing cells reached  $\sim 1000$  mm<sup>3</sup>, every mouse was euthanized and the tumor tissues were removed for subsequent biochemical analyses.

### In vivo Protea-Tac delivery using AAV

When the MCF7 xenograft tumors reached a volume of  $\sim 100$  mm<sup>3</sup>, AAV serotype 2 (AAV2) encoding *survivin*-driven EGFP or scFv(c-Fos)-ADRM1 was administered intratumorally three times per week ( $2 \times 10^9$  vg in PBS). To ensure tumor-specific expression, all transgenes were placed under the control of the 269-bp *survivin* core promoter. When tumors in the PBS- or AAV2-*survivin*-EGFP-treated groups reached  $\sim 600$  mm<sup>3</sup>, mice were euthanized, and tumor tissues were collected for subsequent biochemical analyses. In case of CT26 allograft assay, when the CT26 allograft tumors reached a volume of  $\sim 150$  mm<sup>3</sup>, AAV2 encoding *survivin*-driven EGFP or scFv(c-Fos)-ADRM1 was administered intratumorally one time per week ( $4 \times 10^9$  vg in PBS). When tumors in the PBS- or AAV2-*survivin*-EGFP-treated groups reached  $\sim 1200$  mm<sup>3</sup>, mice were euthanized, and tumor tissues were collected for subsequent biochemical analyses.

### Serum biochemistry and IL-6 analyses

At the experimental endpoint, blood samples were collected and serum was isolated for subsequent analyses. Serum levels of ALT, AST, ALP, UA, CREA, and BUN were analyzed by KP&T (South Korea). Serum IL-6 concentrations were determined using a Mouse IL-6 ELISA Kit (Invitrogen, KMC0061) according to the manufacturer's instructions. Absorbance was measured at 450 nm using a microplate reader, and IL-6 concentrations were calculated from a standard curve.

### Preparation of mRNA

To generate mRNA, a DNA template includes the 5' untranslated region (UTR), 3'UTR, and poly(A) (polyadenylate) tails. The DNA template featured multiple cloning sites containing four distinct restriction sequences, facilitating the precise insertion of target genes, such as EGFP, scFv(c-Fos), PSMD4-scFv(c-Fos), and scFv(c-Fos)-ADRM1. Before in vitro transcription (IVT), the plasmid DNA was completely linearized using the Not I (Enzynomics, Daejeon, South Korea) restriction enzyme. IVT was then conducted at  $37^{\circ}\text{C}$  for 4 hours using the T7 RNA polymerase (Enzynomics, Daejeon, South Korea) and its associated buffer system to produce mRNA from DNA templates. Uridine was entirely replaced with N<sup>1</sup>-methylpseudouridine (TriLink, San Diego, CA, USA) to minimize innate immune activation, and a Cap 1 analog was used (SMARTCAP, ST PHARM, Seoul, South Korea) during the transcription process according to the manufacturer's instructions. Following the reaction, the template DNA was digested with DNase I (Enzynomics, Daejeon, South Korea) at  $37^{\circ}\text{C}$  for 30 min. The synthesized mRNA was precipitated via

lithium chloride and cellulose-based purification to remove double-stranded RNA (dsRNA) contaminants. The final mRNA products were characterized for integrity using 1% denaturing agarose gel electrophoresis, and their concentration and purity were assessed using a NanoDrop 2000 spectrophotometer (TFS, Waltham).

### Formulation of mRNA-LNP

To formulate mRNA-encapsulated LNPs, we used lipids including ALC-0315, 1,2-distearoyl-*sn*-glycero-3-phosphocholine (DSPC), ALC-0159, and cholesterol (Avanti Polar Lipids, USA). For mRNA-LNP formulation, the lipid phase was dissolved in absolute ethanol and mixed with an acidic mRNA aqueous phase (50 mM citrate, pH 4.0) maintaining an N/P ratio of 6. NanoAssemblr Ignite (Precision Nanosystems Inc., Canada) was used for formulating mRNA-LNPs. The solutions were mixed at a 1:3 volume ratio (organic to aqueous solutions) and a total flow rate of 10 ml/min. To ensure complete solvent exchange and particle stabilization, the synthesized mRNA-LNPs were first subjected to a two-step dialysis process. The formulations were dialyzed against Tris buffer for 2 hours at RT, followed by an overnight incubation at 4°C. In accordance with the manufacturer's recommendations, the final mRNA-LNPs were concentrated using an Amicon Ultra-15 centrifugal filter (Merck Millipore, Germany). Dynamic light scattering (DLS) was performed on a Zetasizer Nano ZS (Malvern Panalytical, Malvern, Worcestershire, UK) to determine particle size with all samples diluted in Tris buffer. Furthermore, the encapsulation efficiency was quantified through a Ribo-green assay using the Quant-it Ribo-Green RNA Assay Kit (TFS, Waltham, MA, USA). Total RNA was released via Triton X-100 treatment, enabling the determination of encapsulation efficiency by comparing internalized versus free RNA.

### In vivo Protea-Tac delivery using mRNA-LNPs

When the average tumor volume reached  $\sim 100 \text{ mm}^3$ , the mice were randomly assigned to treatment groups. The mice were treated with a subcutaneous injection of 40  $\mu\text{g}$  of mRNA per dose formulated in LNPs encoding EGFP, scFv(c-Fos), PSMD4-scFv(c-Fos) in a total volume of 50  $\mu\text{l}$ , whereas the control group was injected with 50  $\mu\text{l}$  of PBS via the same route. Treatments were administered every 5 days for a total of three doses. Tumor growth was monitored every 2 to 3 days. Eleven days after the final injection, the mice were euthanized via CO<sub>2</sub> asphyxiation. Blood was collected from the caudal vena cava for serum biochemical analysis, and tumor tissues were harvested for further analyses.

### Densitometry and statistical analysis

Band intensities from immunoblots were quantified by densitometric analysis using the iBright Analysis Software (v5.4.0). Statistical significance between groups was assessed using an unpaired *t* test, one-way analysis of variance (ANOVA), or two-way ANOVA followed by Tukey's and Sidak's multiple comparison tests as post hoc analyses (GraphPad Prism, version 10.4.1). A *P* value of less than 0.05 was considered statistically significant. For TMT-MS, protein abundance data were processed in Perseus (version 1.6.15.0). Reporter ion intensities were log<sub>2</sub> transformed, and width adjustment normalization was applied to equalize interquartile ranges across samples. Principal components analysis was conducted to visualize overall sample variation and to assess the separation between controls and experimental groups. Student's *t* test was performed to identify differentially expressed proteins, applying a Benjamini-Hochberg

FDR threshold of  $<0.05$  and an FC cutoff of  $>1.2$  or  $<0.833$ , corresponding to a 20% increase or decrease in expression ( $|\log_2\text{FC}| > 0.263$ ).

### Supplementary Materials

The PDF file includes:

Figs. S1 to S7

Table S1

Legend for data S1

Other Supplementary Material for this manuscript includes the following:

Data S1

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#### Acknowledgments

**Funding:** This work was supported by the grants from the National Research Foundation of Korea (RS-2025-16652968 and RS-2026-25477259 to M.J.L.), the Ministry of Food and Drug Safety (RS-2025-02213409 to J.-H.N.), and the Samsung Research Funding & Incubation Center of Samsung Electronics (SRFC-MA2202-06 to C.H.L., J.-H.N., and M.J.L.). **Author contributions:** Conceptualization: M.J.L. and J.-H.N. Methodology: S.H.P., Y.J., M.J.L., J.K., D.H., D.J., S.-Y.L., J.-H.N., C.-H.L., and I.B. Investigation: J.Y., S.H.P., Y.J., J.K., D.H., D.J., S.-Y.L., C.-H.L., E.K., and I.B. Resources: J.Y., S.H.P., Y.J., D.H., S.-Y.L., and I.B. Data curation: J.Y., S.H.P., Y.J., M.J.L., J.K., D.H., D.J., S.-Y.L., J.-H.N., C.-H.L., and E.K. Validation: J.Y., S.H.P., Y.J., M.J.L., J.K., D.J., S.-Y.L., J.-H.N., C.-H.L.,

and I.B. Formal analysis: S.H.P., Y.J., J.K., D.H., D.J., S.-Y.L., E.K., and I.B. Visualization: S.H.P., Y.J., J.K., D.J., S.-Y.L., C.-H.L., E.K., and I.B. Project administration: S.H.P., M.J.L., J.-H.N., and C.-H.L. Supervision: M.J.L., J.-H.N., and C.-H.L. Funding acquisition: M.J.L. and J.-H.N. Writing—original draft: J.Y., S.H.P., Y.J., M.J.L., and C.-H.L. Writing—review and editing: S.H.P., Y.J., M.J.L., J.K., D.J., S.-Y.L., J.-H.N., and C.-H.L. **Competing interests:** S.H.P., Y.J., and M.J.L. are inventors on pending patent applications related to this work filed by the Seoul National University: China application no. 202510908779.5, filed on 2 July 2025; European application no. 25187127.3, filed on 3 July 2025; and US application no. 19/011,082, filed on 6 January 2025. The other authors declare that they have no competing interests. **Data, code, and materials availability:** The raw TMT-MS/MS proteomics data have been deposited at the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifier PXD063133 (<https://ebi.ac.uk/pride/archive/projects/PXD063133>). The coding sequences of the Protea-Tac and related chimera constructs used in this study are provided in table S1. All other data and code needed to evaluate and reproduce the results in the paper are present in the paper and/or the Supplementary Materials. Author-generated plasmid constructs and stable cell lines generated in this study can be provided by M.J.L. pending scientific review and a completed material transfer agreement; requests for these materials should be submitted to M.J.L. (minjlee@snu.ac.kr). Source plasmids used to derive author-generated plasmids were obtained from Addgene. Plasmid no. 182090: <https://addgene.org/182090/>, no. 198303: <https://addgene.org/198303/>, no. 198302: <https://addgene.org/198302/>, and no. 166518: <https://addgene.org/166518/>. Commercially available reagents used in this study are listed in Materials and Methods. No custom code or software was generated in this study.

Submitted 9 March 2026

Accepted 7 August 2026

Published 18 September 2026

10.1126/sciadv.aeh0227